

Autonomous Vehicle - Concept Map

Overview & Features of Computer Vision

- Helps the car understand its surroundings
- Uses camera and AI
- Real time processing

Image Sensing & Application

- Cameras (RGB / Infrared)
- Capture real world image
- Convert to digital form



Digital Image Fundamental

- Pixels
- Resolution
- Intensity
- Contrast
- Color Spaces



Application

- Lane detection
- Traffic sign recognition
- Obstacle detection



Sampling & Quantization

- Spatial sampling
- Quantization
- Convert analog to digital



Image Formation Models

- Camera geometry
- Light & illumination
- Object formation



Image Preprocess

- Filtering
- Segmentation

Feature Extr.

- Edge
- Shape
- Texture



Object Recogn.

- Bicycle
- Vehicle
- Traffic Sign



Navigation Safe Driving



Medical Imaging - Concept Map

Overview & Feature of Computer Vision

- * Capture and understand image content
- * Detect patterns and objects
- * Help doctors identify abnormalities

Image Sensing & Application

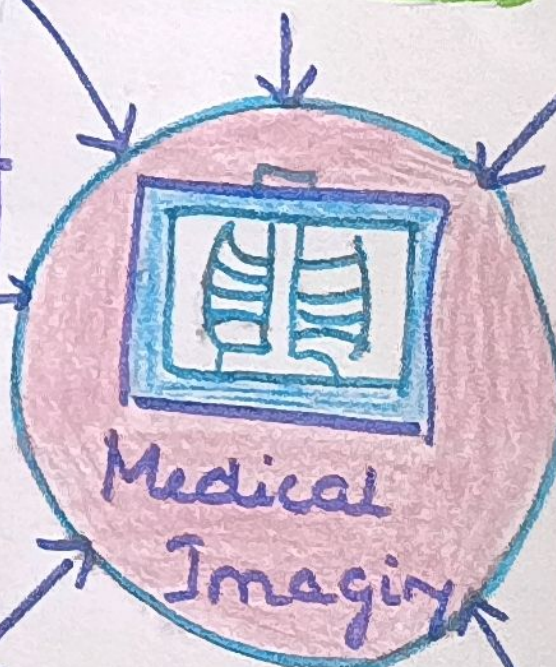
- * X-ray source
- * passes through body
- * Detector Capt Image
- * Conv. - digital

Digital Image Fundamentals

- * Pixels
 - * Resolution
 - * Intensity
 - * Contrast
 - * Image Representation
- 

Application

- * Digital X-ray
 - * CT Scan
 - * MRI
 - * Ultrasound
- 



Sampling & Quantization

- * Spatial Sampling
 - * Intensity Quantization
 - * Convert analog to digital
- 

Levels of CV & Image Processing

- High level
- Mid level
- Low level

Image Formation Mode

- * X-ray
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Image processing

- * Filtering
- * Segmentation

Feature Extraction

- * Edge, Shape, Texture
- 

Image Analysis

- * pattern recognition
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Diagnosis

- * Plan treatment
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